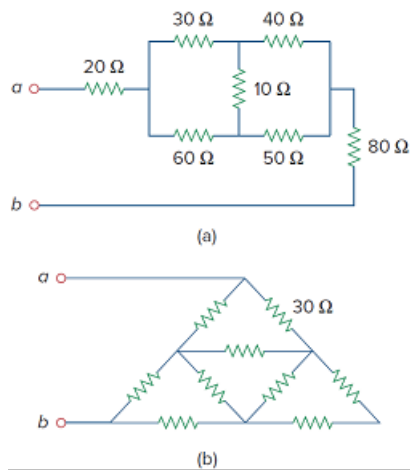


Problem

Obtain the equivalent resistance R_{ab} in each of the circuits of Fig. 2.117. In (b), all resistors have a value of $30\ \Omega$.

Figure 2.117:



Step-by-step solution

Step 1 of 16

(a)

Refer to figure 2.117 (a) in the textbook.

Redraw the circuit as shown in figure 1.

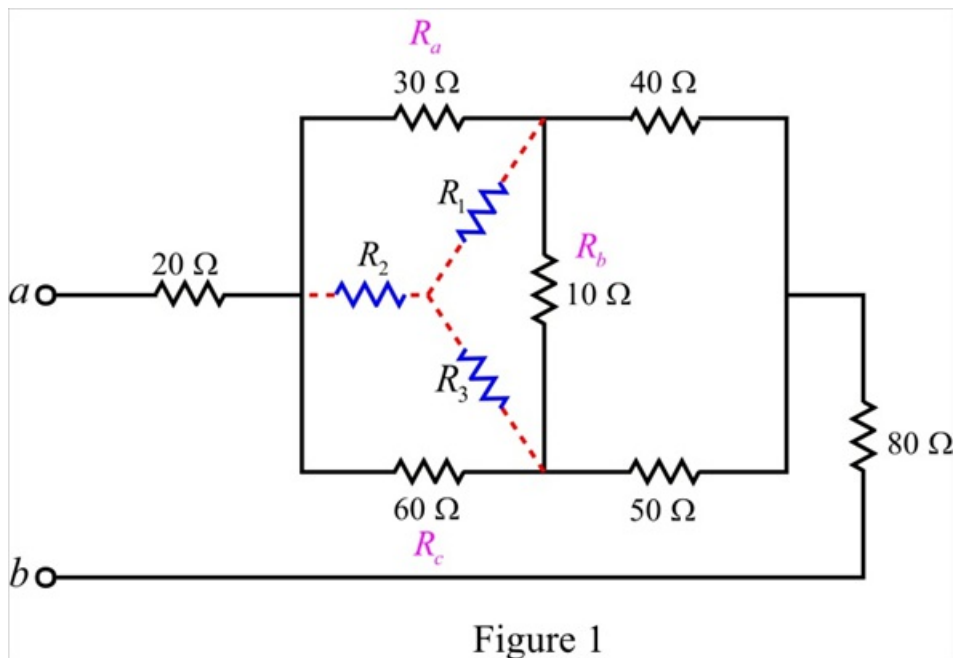


Figure 1

Step 2 of 16

Transform Delta network into Wye network with resistances R_1, R_2 , and R_3 as shown in figure 1.

Calculate the resistance R_1 .

$$R_1 = \frac{R_a R_b}{R_a + R_b + R_c}$$

Substitute 30 for R_a , 10 for R_b , and 60 for R_c .

$$\begin{aligned} R_1 &= \frac{30 \times 10}{30 + 10 + 60} \\ &= \frac{300}{100} \\ &= 3 \, \Omega \end{aligned}$$

Step 3 of 16

Calculate the resistance R_2 .

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

Substitute 30 for R_a , 10 for R_b , and 60 for R_c .

$$\begin{aligned} R_2 &= \frac{30 \times 60}{30 + 10 + 60} \\ &= \frac{1800}{100} \\ &= 18 \, \Omega \end{aligned}$$

Step 4 of 16

Calculate the resistance R_3 .

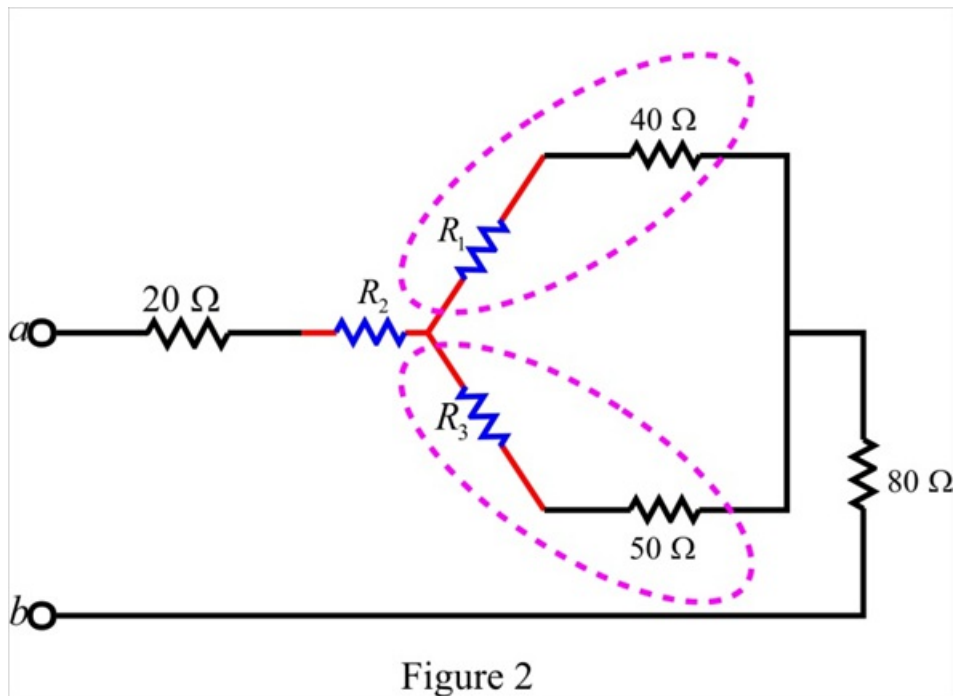
$$R_3 = \frac{R_b R_c}{R_a + R_b + R_c}$$

Substitute 30 for R_a , 10 for R_b , and 60 for R_c .

$$\begin{aligned} R_3 &= \frac{10 \times 60}{30 + 10 + 60} \\ &= \frac{600}{100} \\ &= 6 \, \Omega \end{aligned}$$

Step 5 of 16

Redraw the circuit with resistances R_1, R_2 , and R_3 as shown in figure 2.



Step 6 of 16

Consider the circuit shown in figure 2.

The resistances $(R_1 + 40) \Omega$ and $(R_3 + 50) \Omega$ are parallel to each other.

Calculate their equivalent resistance.

$$R_4 = \frac{(R_1 + 40)(R_3 + 50)}{R_1 + 40 + R_3 + 50}$$

Substitute 3 for R_1 , and 6 for R_3 .

$$\begin{aligned} R_4 &= \frac{(3 + 40)(6 + 50)}{3 + 40 + 6 + 50} \\ &= \frac{43 \times 56}{99} \\ &= \frac{2408}{99} \Omega \end{aligned}$$

Step 7 of 16

Calculate the equivalent resistance R_{ab} .

$$R_{ab} = 20 + R_2 + R_4 + 80$$

Substitute 18 for R_2 , and $\frac{2408}{99}$ for R_4 .

$$\begin{aligned} R_{ab} &= 20 + 18 + \frac{2408}{99} + 80 \\ &= \frac{14090}{99} \\ &= 142.323 \Omega \end{aligned}$$

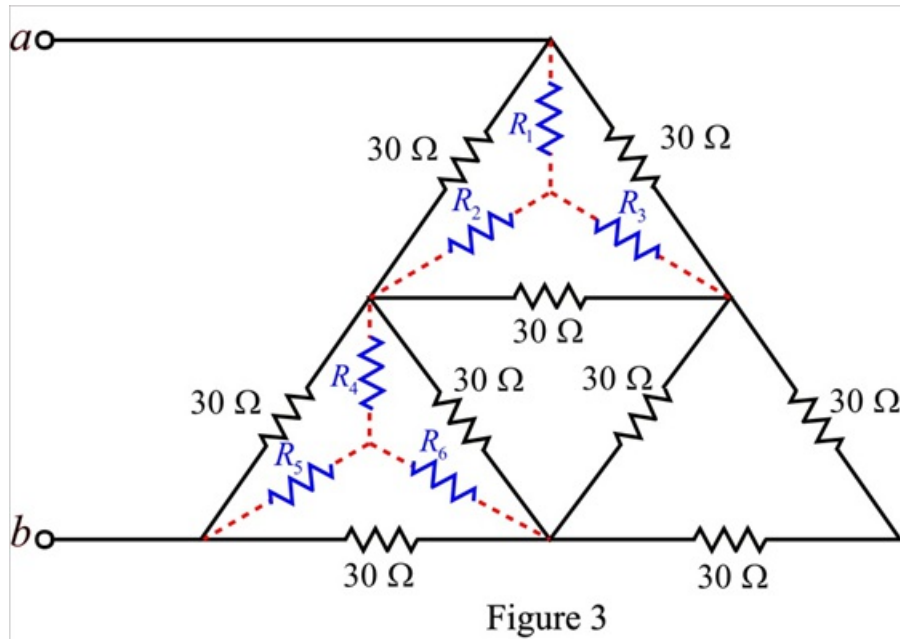
Therefore, the value of equivalent resistance R_{ab} is 142.323Ω .

Step 8 of 16

(b)

Refer to figure 2.117 (b) in the textbook.

Redraw the circuit as shown in figure 2.



Step 9 of 16

Transform Delta networks into Wye networks as shown in figure 3.

Calculate the resistances R_1 , R_2 , and R_3 .

$$R_1 = \frac{30 \times 30}{30 + 30 + 30} \\ = 10 \, \Omega$$

$$R_2 = \frac{30 \times 30}{30 + 30 + 30} \\ = 10 \, \Omega$$

$$R_3 = \frac{30 \times 30}{30 + 30 + 30} \\ = 10 \, \Omega$$

Step 10 of 16

Calculate the resistances R_4 , R_5 , and R_6 .

$$R_4 = \frac{30 \times 30}{30 + 30 + 30} \\ = 10 \, \Omega$$

$$R_5 = \frac{30 \times 30}{30 + 30 + 30} \\ = 10 \, \Omega$$

$$R_6 = \frac{30 \times 30}{30 + 30 + 30} \\ = 10 \, \Omega$$

Step 11 of 16

Redraw the circuit as shown in figure 4.

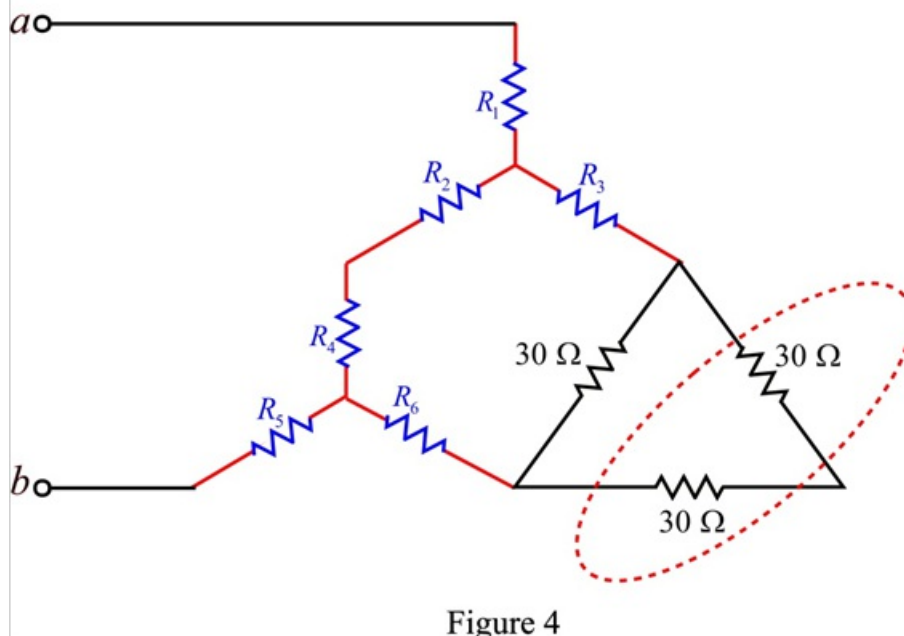


Figure 4

Step 12 of 16

Consider the circuit shown in figure 4.

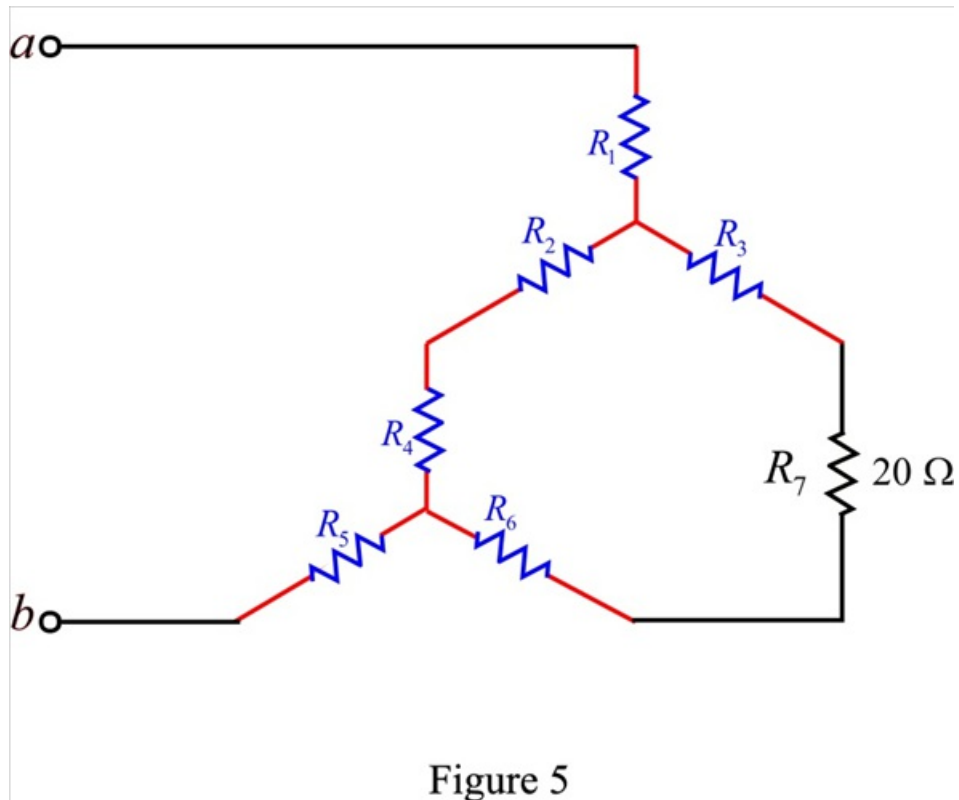
The two resistances each of $30\ \Omega$ are in series and their combination is parallel to a resistance of $30\ \Omega$.

Calculate the equivalent resistance R_7 .

$$\begin{aligned} R_7 &= \frac{(30 + 30)30}{30 + 30 + 30} \\ &= \frac{60 \times 30}{90} \\ &= \frac{1800}{90} \\ &= 20\ \Omega \end{aligned}$$

Step 13 of 16

Redraw the circuit as shown in figure 5.



Step 14 of 16

The resistances $(R_2 + R_4) \, \Omega$ and $(R_3 + 20 + R_6) \, \Omega$ are parallel to each other.

Calculate their equivalent resistance R_8 .

$$R_8 = \frac{(R_2 + R_4)(R_3 + 20 + R_6)}{(R_2 + R_4) + (R_3 + 20 + R_6)}$$

Substitute 10 for R_2, R_3, R_4 , and R_6 .

$$\begin{aligned} R_8 &= \frac{(10+10)(10+20+10)}{(10+10)+(10+20+10)} \\ &= \frac{(20)(40)}{(20)+(40)} \\ &= \frac{40}{3} \, \Omega \end{aligned}$$

Step 15 of 16

Redraw the circuit as shown in figure 6.

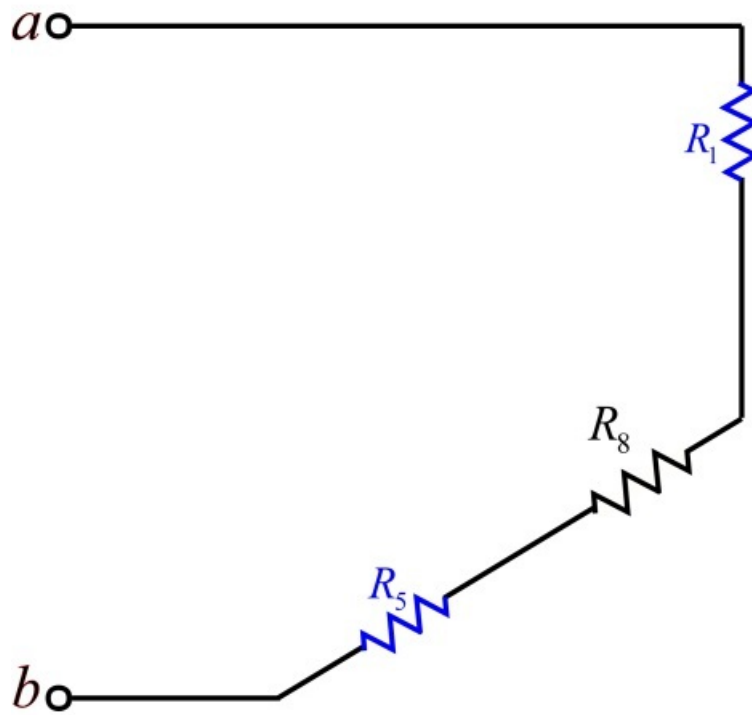


Figure 6

Step 16 of 16

Calculate the equivalent resistance R_{ab} .

$$R_{ab} = R_1 + R_8 + R_5$$

Substitute 10 for R_1 and R_8 ; and $\frac{40}{3}$ for R_5 .

$$\begin{aligned} R_{ab} &= 10 + \frac{40}{3} + 10 \\ &= 33.33 \, \Omega \end{aligned}$$

Therefore, the value of equivalent resistance R_{ab} is $33.33 \, \Omega$.